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# Identification of Fake Products using Blockchain Technology

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***Abstract* - Blockchain, which has recently received a lot of attention, has a wide range of uses. Bitcoin, a widely used cryptocurrency, employs blockchain technology to authenticate transactions without relying on a central structure, thereby solving the problem of double spending. Consequently, the application that uses blockchain technology as its core ensures the authenticity of the data. The research uses a concept that is based on the decentralised blockchain to demonstrate that customers do not primarily rely on stores to evaluate the legitimacy of products. This study describes a decentralised blockchain architecture that is free of counterfeit items so that makers can use it to make real products without having to contact merchants directly, which will result in a significant reduction in the cost of product quality monitoring.**

***Keywords–Blockchain, Legitimate, Counterfeit, Decentralised, IoT***

## I. INTRODUCTION

Blockchain is an innovative technology that might change computing in the future and offer more innovative solutions across several sectors. It can be used just about any place since it is distributed, immutable, and open. Even if the technology has many applications outside of banking, the emergence of cryptocurrencies has increased its appeal [1]. Multiple cryptographically linked blocks are a rough translation of the term "blockchain". The data, the hash of the block before it, and the hash of the data and the preceding hash are the three components that make up a block in a data structure [2]. As a consequence, employing the order of dependence between blocks makes it possible to ensure the integrity of the whole blockchain [3]. The hash of a block will change each time its contents change. The spiral effect that will follow from this will render the hashes of the next blocks incorrect. Transactions on the blockchain cannot be altered as a result.

The sale of counterfeit goods is expanding, which has an impact on the revenue and profits of impacted businesses [4]. This article suggests a blockchain-based technique for avoiding counterfeit goods by confirming the correctness of the product and authenticating actual commodities across the supply chain. Only a small processing fee is required for businesses to avoid purchasing counterfeit items. Though it has been suggested, phoney goods identification technology needs to be improved [5].

Blockchain is a type of distributed ledger technology. When a technical advancement is controlled by several parties, such as a secure database, it is said to be decentralised. This innovative technique was used for the first time by Bitcoin. Before creating a new block, the algorithm that manages the blockchain's block creation process collects and verifies data [6].

Blockchain is a decentralised ledger system that keeps track of information like a secure database that is accessible to many people at once. Only Bitcoin has ever used this cutting-edge innovation [7]. Before creating a new block, the Blockchain's production algorithm collects and verifies data. First, we'll look at how Bitcoin serves as an example of the Blockchain's consensus mechanism. The proof-of-work algorithm is utilised by the blockchain. The blockchain validates transactions through a proof-of-work mechanism [8]. In order to solve a difficult mathematical problem that can be verified quickly, each node competes for processing power to solve the problem. Also, every node in the network functions as a data repository for the blockchain and as a hub for communicating with other nodes. An identical set of blockchain records is stored on every node. Each node will rely on its own information to verify incoming transactions [9]. Then, they are added to the next block along with an offer for the rights to record the transactions in that block.

Trust, which develops when two parties interact frequently with a third party, is the foundation of much economic activity in modern society. Due to the low level of trust between the parties involved, reliance on third-party assurances of trust is common [10]. As a result, it is crucial to zero in on the aspects of blockchain technology that allow it to disrupt the age-old basis of financial transactions. With the help of developing


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blockchain technology, an individual can create his own decentralised data-recording system without the need for a middleman [11].

## II. RELATED WORK

According to Da'u and N. Salim [1], the conventional structure of cloud storage is central, and they explain how a single problem has the potential to put the entire system to a standstill. IPFS, the blockchain used by Ethereum, attribute-based authentication, and a distributed file system are all elements that have been incorporated into the architecture. The problem that cloud servers often produce inaccurate results in conventional storage systems is addressed by the decentralised architecture that is built on the ethereum blockchain. This design offers a keyword search capability that can be performed on the ciphertext.

A framework is proposed by M. Xu and S. Liu [2] to verify the originality of media files before they are made available online. Technologies like blockchain and IPFS are used in the framework. The solution relies on the uniqueness, authenticity, and integrity of digital books, but the architecture, design, logic, and smart contract code are meant to be modified and applied to bring these qualities to all other forms of digital goods.

S.Paul, J.I.Joy et al. [3] proposed using blockchain technology to provide security information about the food supply chain, in contrast to the traditional supply chain. By enabling blockchain to track, monitor, and audit the food supply chain as well as help producers properly record transactions, the suggested design addresses the shortcomings. The suggested strategy has not been tried out in actual operations; it is merely a theoretical idea.

HACCP was used by Li ST et al. [4] to describe the functioning of blockchain in the food production chain. The method established a state-of-the-art, decentralised tracking system using IoT and blockchain technologies while also resolving the blockchain's general scalability concerns. When implemented, the customised system would give all stakeholders in the food production chain access to up-to-the-minute information on product safety. The framework has the potential to greatly improve food supply chain quality and transparency, which in turn could lead to greater food safety and boost consumer trust in the food industry.

Using blockchain technology, Si Chen, Rui Shi, et al. [5] showed how to increase trust in news spread via social media. The concept proposed a cutting-edge, decentralised tracking system that combines social media and block chains. After that, you can request the user ID. The transaction will notify all nodes whenever new data is generated. The framework will only examine "viral" news stories that have attained a predetermined level of attention. The chain will receive the delivery of information. Initially, this news will not have a rating. The ranks of users will be displayed in the news, and the ratings of validators will accumulate over time. This rating indicates the dependability and authenticity of the news item.

Mitsuaki Nakasumi [6] proposed a blockchain-based remedy for supply chain issues such as Double Marginalisation and Information Asymmetry. Systems for managing the supply chain help businesses plan by exchanging, analysing, and supporting information. Due to unequal knowledge between organisations, which disrupts the planning algorithms, they are not founded on actual facts. On the other side, information sharing between producers, suppliers, and customers becomes essential to guaranteeing market responsiveness. The problem of double marginalisation in supply chain management is pervasive and important. Double marginalisation effects are shown to produce supply insufficiencies in decentralisation under wholesale pricing contracts when both predictable and unpredictable demand is present.

In a recent study, Khalil G. et al. [7] report the use of a frequency identification tag in the detection of counterfeit goods. This method uses radio frequency identification tags on products to conduct a review based on a test topic for counterfeit goods. While remote sensor networks and radio frequency identification (RFID) tags are frequently used in this system for remote development and offer a range of potential future uses, RFID tags, like actuators, need a control signal and a power source. Instead of employing laser light from verified tags to read areas and physical proof of tagged objects, RFID collects and stores data using low-frequency radio waves. This technology is used to handle the collection of data in a cellular environment or a central warehouse. Before transmitting them to the RFID tag, the transceiver performs a scan of the radio waves. After that, the RFID reader sends separate data to a microchip that is very small.

## III. PROPOSED SYSTEM

Because most consumers purchase from remote locations, there is a risk of receiving counterfeit or phoney goods. This bogus product harms both the client and the company's reputation. They will suffer significant losses as a result of this predicament [12]. Before dealing with this issue, there is no correct solution. There is no assurance mechanism or a suitable technique to detect counterfeit items from genuine ones since barcodes are easily reproduced. Blockchain is the most promising developing technology in recent years that can assist in the resolution of this sort of problem. Blockchain technology may be used to monitor and track transported items, ensuring that users only receive the correct product [13]. The major goal this study is to increase openness about the goods during client purchases and make it easier for customers to determine if the product they were purchasing was authentic or counterfeit.

This study uses blockchain technology to identify fake products. The initial stage is to connect all manufacturers to the blockchain network and collect their primary product information. Product verification is done by signing up and giving the right username and password. The main owner of the item will be the person who made it. The maker will inquire with the management about connecting the product to the network, and the manager will provide the maker with the QR code for the product, as depicted in Fig. 1.


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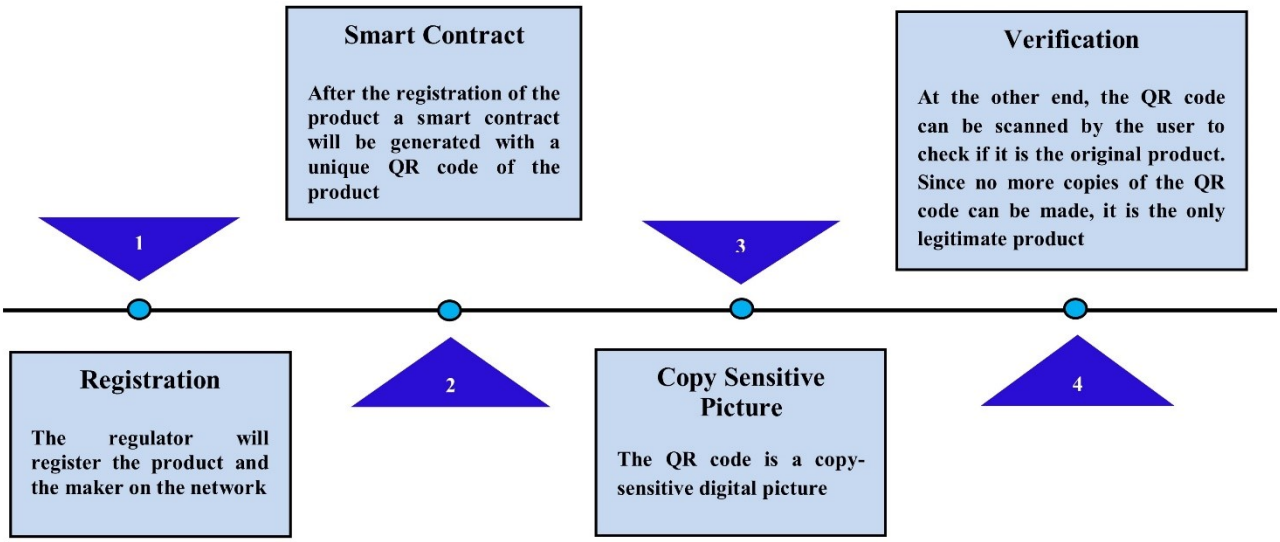


Fig. 1. Flow of Verification

If the application is from a legitimate manufacturer, the regulator will add the product and maker to the network. Once the product is registered on the network, it will generate a smart contract containing the product's unique QR code and encrypted text data. The QR code contains a copy sensitive digital image to prevent it from being copied [14]. The manufacturer will then ship the goods to the distributor, and the status will be changed to shipment; ownership will not change until both parties approve a purchase and sale request for the product. When both parties agree to a joint venture, ownership in the blockchain network is instantly transferred in the form of a smart contract upon receipt of payment. At this point, clients will be given the android app, and consumers will be able to scan the QR code assigned to the object using the android app. The scanner scans the product and uses the specified method to extract the encrypted text, providing information about the current maker and owner of the goods and allowing the user to decide whether or not to purchase the item.

## IV. METHODOLOGY

As depicted in Fig. 2, the traceability system architecture was envisioned as layered software consisting of three modules: blockchain, gateway, and device. The device module first transforms the sensed values for the appropriate smart contract into blockchain transactions. In order to provide the necessary network connectivity, the gateway module thus functions as a straightforward and transparent redirect component between the device and the blockchain modules. The blockchain module also provides the twin contract and app contract forms of smart contracts. The app contract performs a separate traceability system mechanism, such as the transfer of ownership between stakeholders, while the twin contract represents IoT devices [15]. As a result, app contracts and twin contracts are capable of direct communication and are regarded as reliable oracles for the blockchain system.

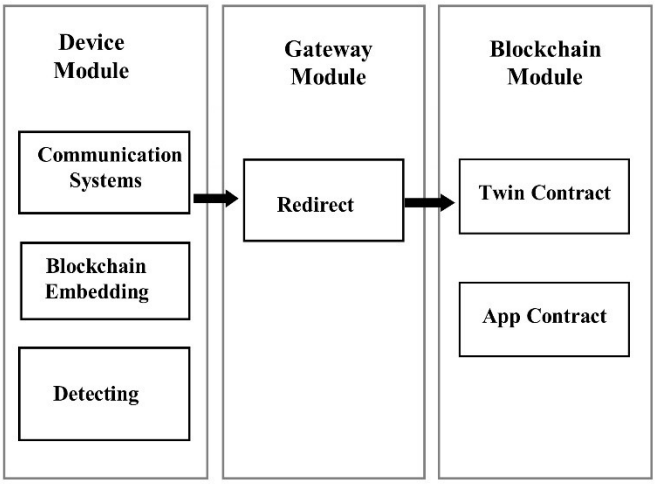


Fig. 2. Traceability system architecture

The details of each module's implementation are provided in the paragraphs that follow, as shown in Fig. 3.


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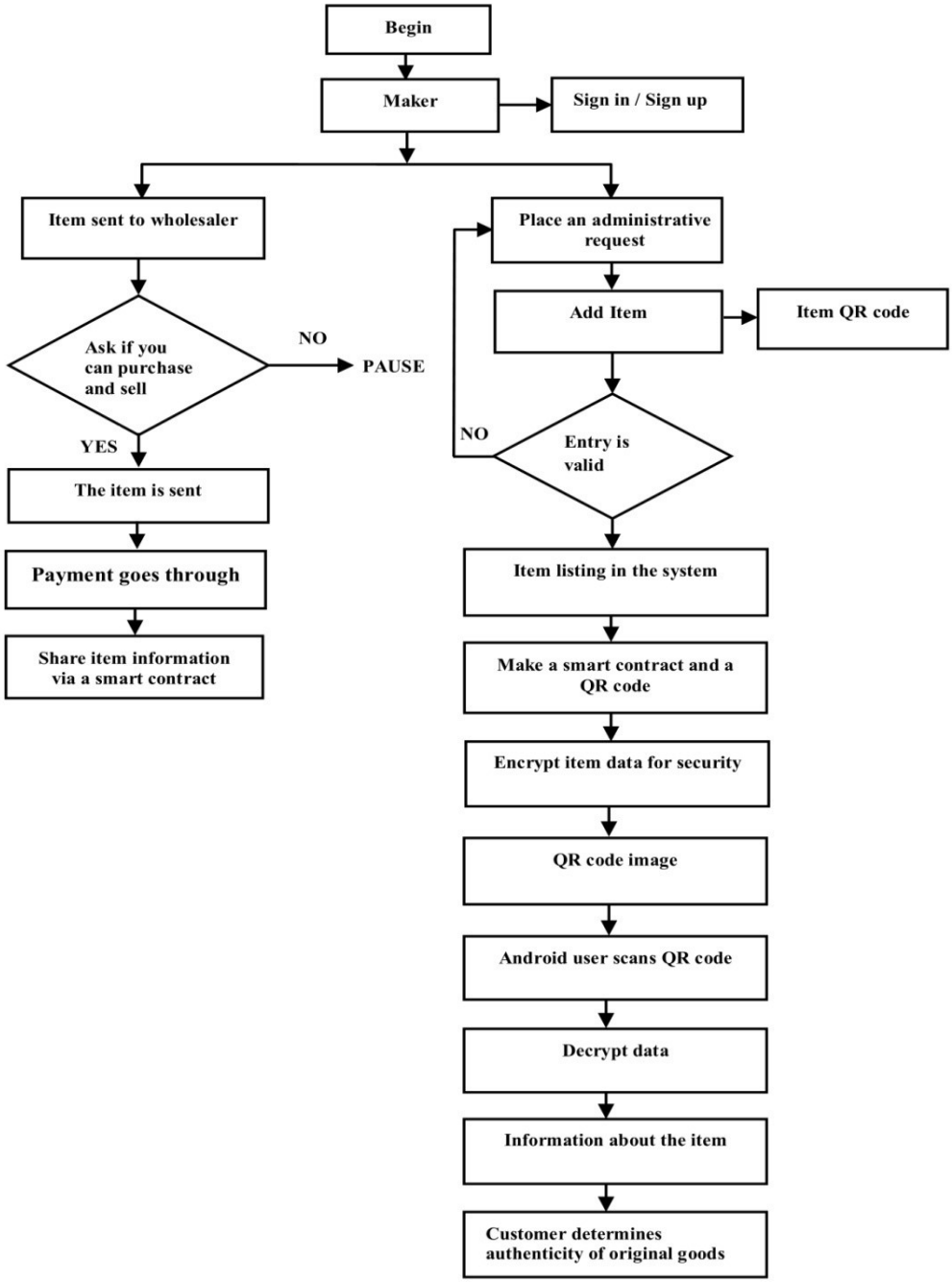


Fig. 3. Schematic representation of the blockchain system

## V. RESULTS AND DISCUSSIONS

Following the rapid growth of online trading apps, several studies have been done with the goal of spotting bogus items. The suggested solution demonstrates how blockchain technology may be efficiently used to limit counterfeit items that circulate in the outside world and harm a company's reputation among consumers. By implementing this, we believe if human errors are neglected, we can bring down the distribution of counterfeit items to nearly 0%, which is 3.3% according to current World Trade Reports. The goal is for anybody to identify counterfeit items without relying on any centralised method or person. Further assessments will be required to ascertain the precise consequences of the system. However, the fate of the system may depend mostly on beginning a blockchain and writing data into a block that may be reviewed or extracted over an extended period of time [16].

As the value of cryptocurrencies and other blockchain-based NFTs rapidly rises, the cost of beginning each block will be determined entirely by its individual rate, and this will have a significant impact on the block's design. As a result, we need a system where the price of the blockchain doesn't change the


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trajectory of the project as a whole if we are going to keep going in this direction.

## VI. CONCLUSION AND FUTURE SCOPE

This research presents a fully operational anti-product counterfeiting system based on blockchain technology. Users of our system can shop with confidence, knowing they are not buying fakes, all for the low price of a single transaction. The architecture enables manufacturers to keep connected product data on blockchain, which may be accessed by any user. Additionally, the seller's total revenue and the number of remaining items are displayed. The user can also use the system's functionality to do immediate vendor-side verification. Digital signatures are employed in this approach to verify the identification. The amount of effort required to successfully run an application that is based on Ethereum is directly proportional to the amount of complexity contained within the program's source code. Future work on this technology may demonstrate code simplification. The client can rest easy knowing that the distributed software won't necessitate any more resources because of the code's simplicity and lack of redundancy. Thus, these vulnerabilities will be fixed in future projects using the revised system.

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